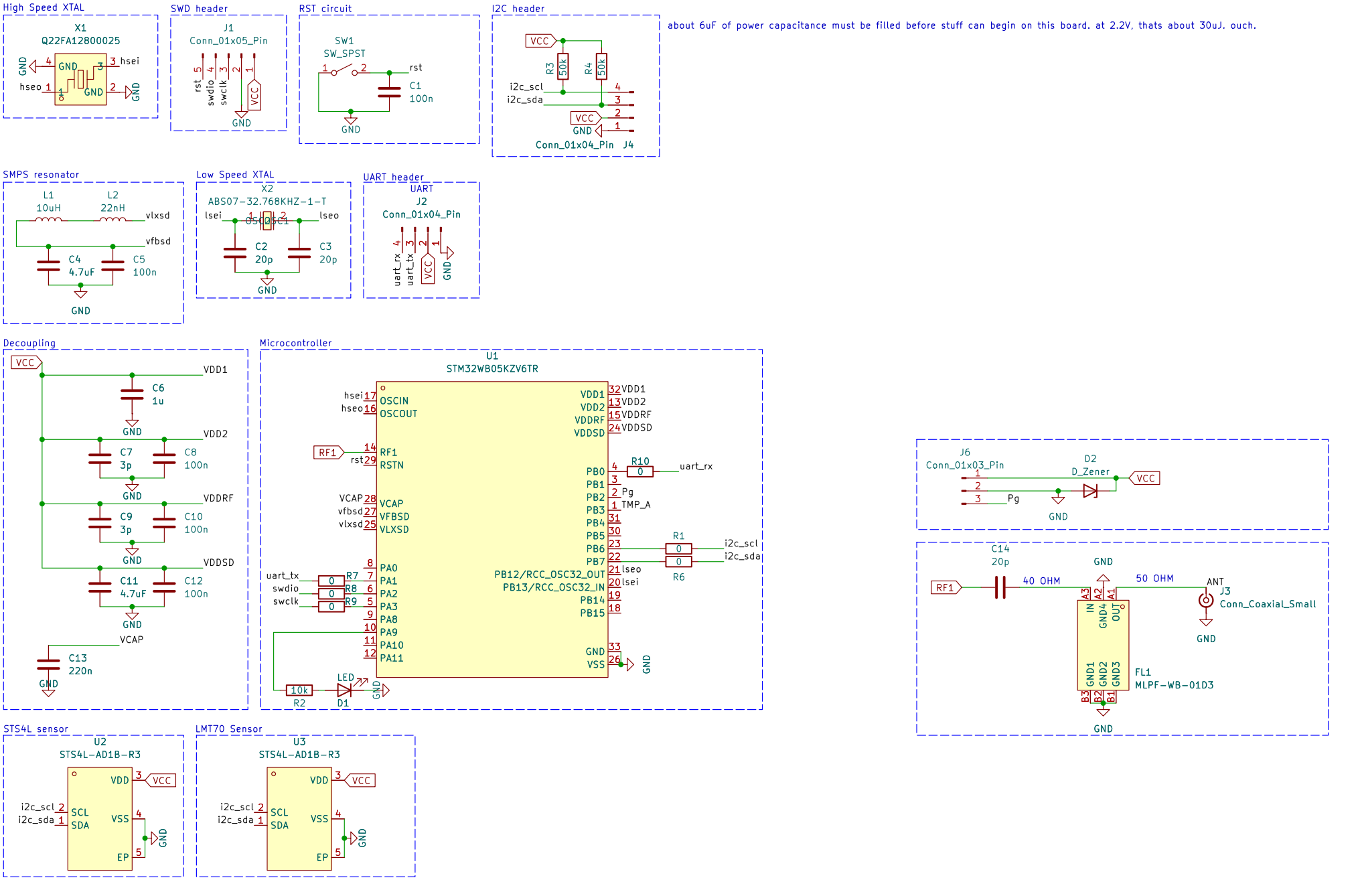

High Speed XTAL
X1
Q22FA12800025
GND
hsei
hseo
SWD header
J1
Conn_01x05_Pin
rst
swdio
swclk
VCC
GND
RST circuit
SW1
SW_SPST
rst
C1
100n
GND
I2C header
VCC
R3
50k
R4
50k
i2c_scl
i2c_sda
VCC
GND
Conn_01x04_Pin
J4
about 6uF of power capacitance must be filled before stuff can begin on this board. at 2.2V, thats about 30uJ. ouch.
SMPS resonator
L1
10uH
L2
22nH
vlxsd
vfbsd
C4
4.7uF
C5
100n
GND
Low Speed XTAL
X2
ABS07-32.768KHZ-1-T
lsei
lseo
C2
20p
C3
20p
GND
UART header
UART
J2
Conn_01x04_Pin
uart_rx
uart_tx
VCC
GND
Decoupling
VCC
VDD1
C6
1u
GND
VDD2
C7
3p
C8
100n
GND
VDDRF
C9
3p
C10
100n
GND
VDDSD
C11
4.7uF
C12
100n
GND
VCAP
C13
220n
GND
Microcontroller
U1
STM32WB05KZV6TR
hsei
hseo
OSCIN
OSCOUT
RF1
rst
RF1
RSTN
VCAP
vfbsd
vlxsd
VCAP
VFBSD
VLXSD
PA0
PA1
PA2
PA3
PA8
PA9
PA10
PA11
uart_tx
swdio
swclk
R7
R8
R9
0
LED
10k
R2
D1
GND
VDD1
VDD2
VDDRF
VDDSD
VDD1
VDD2
VDDRF
VDDSD
PB0
PB1
PB2
PB3
PB4
PB5
PB6
PB7
PB12/RCC_OSC32_OUT
PB13/RCC_OSC32_IN
PB14
PB15
R10
uart_rx
Pg
TMP_A
R1
R6
i2c_scl
i2c_sda
lseo
lsei
GND
VSS
J6
Conn_01x03_Pin
D2
D_Zener
VCC
Pg
GND
C14
20p
RF1
40 OHM
GND
50 OHM
ANT
J3
Conn_Coaxial_Small
IN
GND4
OUT
GND1
GND2
GND3
FL1
MLPF-WB-01D3
GND
STS4L sensor
U2
STS4L-AD1B-R3
VDD
VSS
EP
SCL
SDA
i2c_scl
i2c_sda
VCC
GND
LMT70 Sensor
U3
STS4L-AD1B-R3
VDD
VSS
EP
SCL
SDA
i2c_scl
i2c_sda
VCC
GND